

Preme Malaviphusit

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EDUCATION

New York University, Tandon School of Engineering – New York, NY

Expected May 2028

Bachelor of Science in Electrical and Computer Engineering | Mathematics Minor | GPA: 3.73

- Related Coursework: PCB for Embedded System, Digital Logic, Circuits, Electronics, Data Structure and Algorithms
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EXPERIENCE

NYU General Engineering Department – Brooklyn, NY

Aug 2025 - Present

Head of Prototyping, Teaching Assistant

- Managed Prototyping Lab operations for 300+ students across soldering, embedded systems, CAD, and 3D printing
- Led weekly hands-on labs for 16 students, spanning circuitry, embedded coding, CAD prototyping, and data analysis
- Mentored 4 project teams through full hardware-software integration, resulting in 100% on-time prototype delivery

Flexible AI-Enabled Mechatronic Systems Lab – Brooklyn, NY

Sep 2025 - Dec 2025

Computer Vision Researcher

- Developed a kidney stone detection pipeline using TensorFlow and PyTorch for real-time endoscopic imaging
- Generated 3D volumetric heat maps in slicer software to guide physician decision-making in laser ablation procedures

Metropolitan Electricity Authority – Bangkok, Thailand

Jun 2023 - Jul 2023

Electrical Engineering Intern

- Contributed to MEP and fire protection plan for the Thai Parliament substation using AutoCAD and DIALux
 - Automated fire alarm sound transmission analysis in Excel, cutting manual design review time by 90%
 - Inspected 5+ electrical systems across solar, EV, and hospital infrastructure for performance and compliance
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PROJECTS

Shark Fin Rev 1.0 Avionics Board | NYU ARC Robotics | Altium Designer, LTSpice

Jan 2026 - Present

- Designed a miniaturized PCB with STM32H743VIH6, IMU, and DCMI camera for a guided projectile stabilization
- Architecting dual-input USB-C/LiPo power mux with 3.3V LDO and BGA fanout routing for compact layout

NYU Rogue Aerospace | Electrical Engineer | GPS, RF Telemetry Integration

Jan 2026 - Present

- Built payload electronics for NYU's first hybrid rocket, deploying water via EasyMini apogee detection at ~1,900 ft
- Configured XBee radio modules to establish two-way wireless communication between ground and rocket systems

Custom STM32F405 Development Board | Altium Designer

Jan 2026 - Mar 2026

- Designed a 4-layer STM32F405 PCB in Altium with SPI, I²C, UART, ADC interfaces and USB-C ESD protection
- Implemented 36V XT30 to 5V buck and 3.3V LDO power architecture for stable MCU operation under variable load
- Integrated SWD debug interface with differential pair routing to ensure high-speed signal integrity

Lower Body Exoskeleton | Fusion360, Arduino (C/C++)

Feb 2025 - Apr 2025

- Engineered a lower-limb exoskeleton with encoder-based closed-loop control via L298N H-bridge, achieving <5° joint overshoot for ACL rehabilitation
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LEADERSHIP / EXTRACURRICULAR

Secretary, NYU Society of Asian Scientists and Engineers

May 2026 - Present

Vice President, NYU Society of Women Engineers | Outstanding Event Management Award

Oct 2025 - Present

President, NYU Thai Student Association | Represented NYU at 6,000+ attendee university fair

Jul 2025 - Present

SKILLS

Hardware/PCB: Altium, LTSpice, ModelSim, Vivado, Soldering, Oscilloscope, Multimeter, Power Supplies

Embedded Systems: Arduino, ESP32, STM32, Raspberry Pi

Programming: Python, C/C++, Java, MATLAB, Verilog, HTML/CSS

Mechanical/CAD: Fusion360, SolidWorks, OnShape, AutoCAD